



PLEURAL DECOMPRESSION

Effective: January 1, 2026
Replaces: February 12, 2015

1. Purpose

To describe the procedure of pleural decompression in the prehospital setting. Pleural decompression is a rarely used and highly invasive procedure that is reserved for patients that display rapidly deteriorating vital signs and are suspected of having a tension pneumothorax.

2. Indications

- 2.1. Pleural decompression shall be performed, by ALS providers, when **tension pneumothorax** is suspected due to traumatic injury and the presence of:
 - 2.1.1. Absent or diminished lung sounds on affected side **AND**
 - 2.1.2. Hypotension
 - 2.1.2.1. Adults: Systolic blood pressure (SBP) less than 90 mmHg
 - 2.1.2.2. Pediatrics: SBP less than age-appropriate value (700-S05) or length-based resuscitation tape
 - 2.1.3. **AND** one or more of the following:
 - 2.1.3.1. Altered mental status
 - 2.1.3.2. Adults: Severe respiratory distress, respiratory rate greater than 30/min or less than 10/min
 - 2.1.3.3. Pediatrics: Respiratory rate greater than age-appropriate value (700-S05) or length-based resuscitation tape
 - 2.1.3.4. Severe hypoxia, with SpO2 less than 90%
 - 2.1.3.5. Cool, pale, moist skin

3. Procedure

- 3.1. Choose site for placement of catheter:
 - 3.1.1. 2nd or 3rd intercostal space at the midclavicular line
 - 3.1.2. 4th or 5th intercostal space at the midaxillary line
- 3.2. Prepare the site with betadine/skin prep solution
- 3.3. At a 90° angle insert a 14 g catheter into the skin over the superior border into the pleural space until air escapes or a distinctive “give” is felt
 - 3.3.1. Avoid the inferior side of the rib to limit injury to neurovascular bundle
- 3.4. Remove the needle, leaving plastic catheter in place
- 3.5. Secure catheter and place chest seal over catheter
- 3.6. Reassess lung sounds and vital signs, additional pleural decompressions may be needed

4. Reference

